Probability

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Probability Basics

1. 1/6

2. 1/2 (odd numbers 1, 3, 5, 7 out of 8 sections)

3. E) 1

4. 0.6

5. impossible, unlikely, evens, likely, certain

6. 50%

7. C) Evens

8. 1 (it is certain)

9. Heads, Tails

10. 3/10

11. A, C and E (0, 0.5 and 1 are valid probabilities; 1.2 and −0.3 are not, as probabilities must lie between 0 and 1 inclusive)

12. B) 1/3 (rolling a 5 or 6 is 2 out of 6 outcomes)

13. 15%

14. 1, 2, 3, 4, 5, 6

15. 3/4 (P(red) = 1/4, so P(not red) = 1 − 1/4 = 3/4)

Applying Probability

16. 5/10 = 1/2

17. 0.6

18. D) 0.6

19. HH, HT, TH, TT; P(one head and one tail) = 2/4 = 1/2

20. 60%

21. C) 3/4

22. 1/3 (3 of the remaining 9 counters are red)

23. A, B, C and D are true (E is false — among 1–6 only the number 5 is a multiple of 5, so P = 1/6, not 1/2)

24. 3/5

25. 3/10

26. 28 blue counters (P(red) = 0.3, so red = 0.3 × 40 = 12; blue = 40 − 12 = 28)

27. A) 1/25

28. 1/3 (6/10 × 5/9 = 30/90 = 1/3)

29. 0.025

30. A, B, D and E are true (C is false — P(green) = 3/10 but P(blue) = 4/10, so they are not equal)

Probability Challenge

31. 5 green (red = 1/2×20=10; blue=1/4×20=5; green=20−10−5=5)

32. B) 5/36 (5 of the 36 equally likely outcomes give a total of 8: (2,6),(3,5),(4,4),(5,3),(6,2))

33. 1/15 (first blue: 2/6=1/3; second blue: 1/5; 1/3×1/5=1/15)

34. 1/4 (P(even)=1/2 and P(heads)=1/2; these are independent, so 1/2×1/2=1/4)

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35. A, B, C and E are true (D is false — 1, 4 and 9 are square numbers out of 10, so P=3/10, not 1/5)

36. C) 1/4 (6/9 × 3/8 = 18/72 = 1/4)

37. 1/16 (1/4 × 1/4)

38. 1/6 (whatever the first roll shows, there is a 1 in 6 chance the second roll matches it)

39. 4/5 (originally 4 red and 12 green out of 16; after removing one red, 3 red and 12 green remain out of 15 counters, so P(green) = 12/15 = 4/5)

40. A, B, C and D are true (E is false — P(no tails) = P(HH) = 1/4, not 1/2)

41. 1/2 (P(both odd) = 1/2×1/2=1/4; P(both even)=1/4; 1/4+1/4=1/2)

42. B) 3/5 (P(same colour) = P(both red)+P(both blue) = 3/10+1/10=2/5, so P(different colours)=1−2/5=3/5)

43. 1/4 (1/2 × 1/2)

44. 3/4 (1 − 1/4)

45. 1/3 (of the 9 equally likely outcome pairs, 3 give a total of at least 5: (2,3), (3,2) and (3,3))

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Homework Worksheet — Answers

Probability Basics

1. 1/6

2. 2/5 (even numbers 2 and 4 out of 1, 2, 3, 4, 5)

3. A) 0

4. 0.25

5. impossible, unlikely, evens, likely, certain

6. 50%

7. D) Likely

8. 0 (it is impossible)

9. 1, 2, 3

10. 1/10

11. A, C and E (1, 0.75 and 0 are valid probabilities; −0.5 and 2 are not)

12. B) 1/3 (rolling a 1 or 2 is 2 out of 6 outcomes)

13. 85%

14. HH, HT, TH, TT

15. 2/5 (2 red sections out of 5)

Applying Probability

16. 4/10 = 2/5

17. 0.375

18. C) 0.45

19. HH, HT, TH, TT; P(at least one tail) = 3/4

20. 60%

21. B) 1/2

22. 2/7 (2 of the remaining 7 counters are purple)

23. A, B, C and D are true (E is false — only 5 is a multiple of 5 among 1–8, so P = 1/8, not 1/4)

24. 2/5

25. 1/5

26. 11 yellow (P(green) = 0.45, so green = 0.45 × 20 = 9; yellow = 20 − 9 = 11)

27. A) 1/16

28. 2/9 (5/10 × 4/9 = 20/90 = 2/9)

29. 0.02

30. A, B, C and D are true (E is false — red and green together make 5 of the 10 counters, so P = 0.5, not 0.6)

Standard / Advanced

31. 10 red (2 out of every 6 is red, i.e. 1/3; 1/3 × 30 = 10)

32. C) 8 (2 × 2 × 2)

33. 1/3 (multiples of 3 from 1–9: 3, 6, 9 — that is 3 of the 9 sections, so P = 3/9 = 1/3)

34. 50% (P(red marble) = 1/4 = 25%; doubled = 50%)

35. 7, with probability 6/36 = 1/6

36. 3/4 (red is 1 part of 4, so P(red) = 1/4 and P(not red) = 3/4)

37. A) 1/3 (6/10 × 5/9 = 30/90 = 1/3)

38. 0.1 (0.5 × 0.2)

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39. C) 24 (P(blue) = 1 − 1/4 = 3/4; total = 18 ÷ 3/4 = 24)

40. 1/8 (1/2 × 1/4)

41. 1/2 (an odd total needs one odd and one even score; P = 2 × (1/2 × 1/2) = 1/2)

42. A, B, C and D are true (E is false — P(both red) = 1/15 but P(both blue) = 7/15, so they are not equal)

43. 1/12 (only the pairs (5,6), (6,5) and (6,6) give a product over 25 — 3 of the 36 outcomes)

44. 10 red (red:blue = 2:1, so red = 2/3 × 15 = 10)

45. 1/5 (there are 15 ways to choose 2 tickets from 6, and 3 ways to choose 2 even tickets from the 3 even ones; 3/15 = 1/5)

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